Dynamic Soulbound Reputation NFT System

Verity Reputation Passport

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This report describes the problem, blockchain rationale, system architecture, smart contract, implementation stack, and demonstration links for the project.

Problem definition

Achievements are spread across certificates and institutional platforms. Verifiers must check multiple sources, and digital credentials can be forged, lost, or presented without a clear way to confirm whether they remain valid. This makes reputation difficult to carry across organizations.

The project provides a wallet-bound passport where approved issuers publish credentials and verifiers check issuer, status, score, dates, and an evidence fingerprint. Expired or revoked credentials remain in the audit history but stop contributing to the active score.

Need for blockchain

A shared EVM ledger gives holders, issuers, and verifiers a common record without requiring them to trust one institution’s private database. Contract rules limit credential issuance to approved issuers, prevent passport transfers, and make changes auditable. Sensitive evidence files stay off-chain; the contract stores their SHA-256 fingerprints.

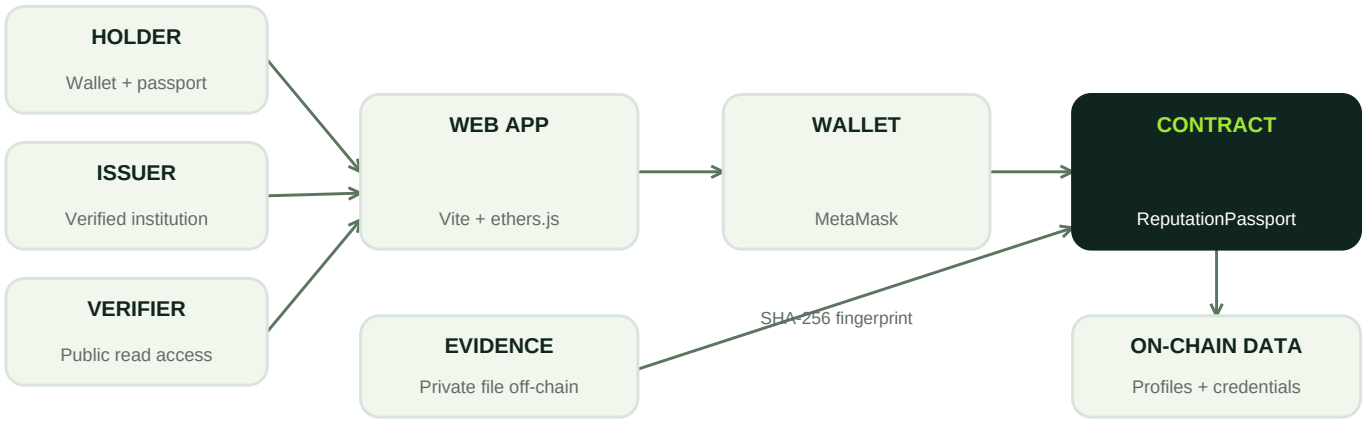
Project links

YouTube upload link (provided Drive folder): Open the supplied video folder

Project launch: https://credsoul.vercel.app

The public frontend loads. Its current configuration reports that the smart contract is not loaded, so on-chain actions require a public-network contract deployment and matching frontend configuration.

Detailed architecture diagram



The holder, issuer, administrator, and verifier use the web app for role-specific actions. MetaMask supplies wallet identity and transaction signatures. The ReputationPassport contract enforces permissions and stores passport and credential state. Evidence files remain private; only their fingerprints are written to the chain.

Blockchain components

- **Wallet:** MetaMask or another EIP-1193 provider identifies the account and signs authorized transactions.
- **Smart contract:** Solidity contract enforces issuer access, one passport per wallet, credential lifecycle rules, and non-transferability.
- **Ledger and events:** Contract state and events record issuance, revocation, and other changes for later verification.
- **Evidence integrity:** SHA-256 hashes let a verifier compare a submitted document to its recorded fingerprint without publishing the document itself.

Smart contract explanation

The ReputationPassport contract mints one non-transferable ERC-721 passport per wallet. An approved issuer can create a student-owned passport with `createProfileFor(recipient)`. Issuers can record credentials with issuer, category, score, dates, and evidence hash. The administrator controls issuer approval.

Credential status is checked against expiry and revocation. `reputationScore()` and `categoryScore()` derive scores from active credentials. `verifyEvidence()` compares evidence fingerprints. Transfer and approval paths revert, and duplicate passport creation is rejected.

Software used

Area	Software / technology
Contract	Solidity, OpenZeppelin Contracts
Development and tests	Hardhat, Node.js, npm
Frontend	Vite, JavaScript, CSS
Wallet and contract access	MetaMask, ethers.js
Authentication	Clerk.js

AI models used

None. Credential issuance, verification, and reputation scoring are implemented with application code and smart-contract rules.

Sample screenshots

Use Demo mode in the live web interface to view the passport dashboard, issuer console, credential list, and credential verification view. The current production frontend does not have a contract configured, so screenshots should show the demo interface and must not imply successful public-chain transactions.

Suggested screenshot set for the submission:

- Passport dashboard with sample score and identity card
- Credential list showing active, expired, and revoked states
- Issuer console showing the credential issuance workflow
- Credential verification result showing issuer, status, and evidence hash

Conclusion

The project demonstrates a wallet-bound, non-transferable reputation passport with issuer-controlled credentials and expiry-aware scoring. On-chain records make credential status and history verifiable, while evidence files remain off-chain. The frontend is publicly reachable; completing live-chain verification requires deploying the contract to a public network and updating the frontend configuration.

Verification performed: Solidity compilation, 17 contract tests, and the Vite production build all passed. The build reports a large JavaScript chunk warning; the output was produced successfully.